

FINAL EXAM

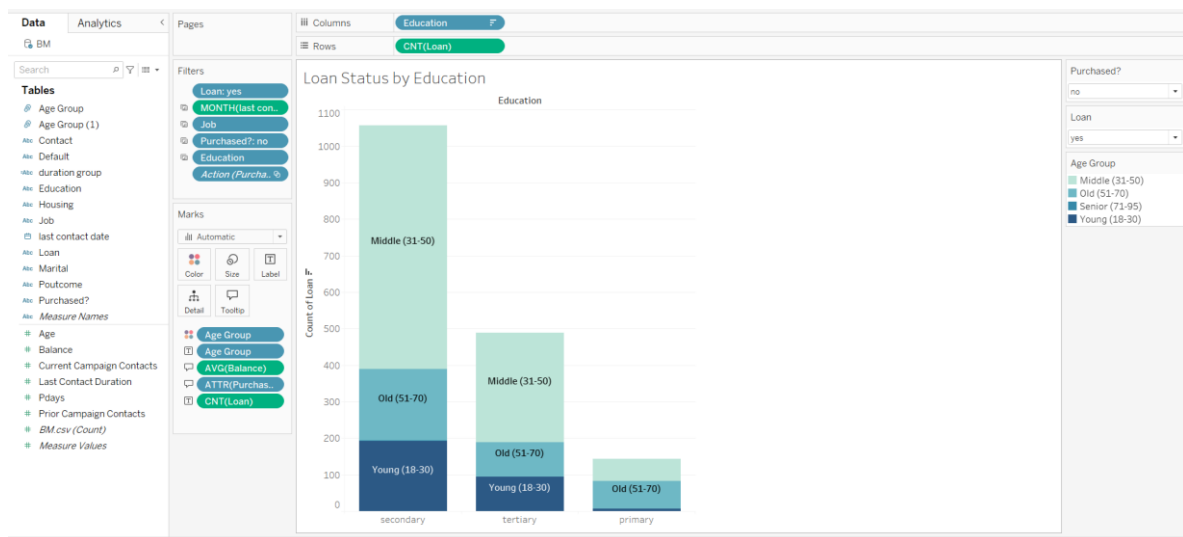
BANK MARKETING ANALYSIS

OVERVIEW

The primary audience for this analysis is marketing managers and data analysts. These professionals seek to understand customer behaviour, enhance engagement strategies, and improve campaign effectiveness.

Our analysis visualizes and interprets data on loan statuses, campaign engagement, and demographic factors related to purchasing behaviour. By using visual tools such as bar charts, line charts, and heatmaps, we aim to uncover patterns and trends that can guide strategic decisions. This approach helps to identify key insights that can drive better marketing and campaign outcomes.

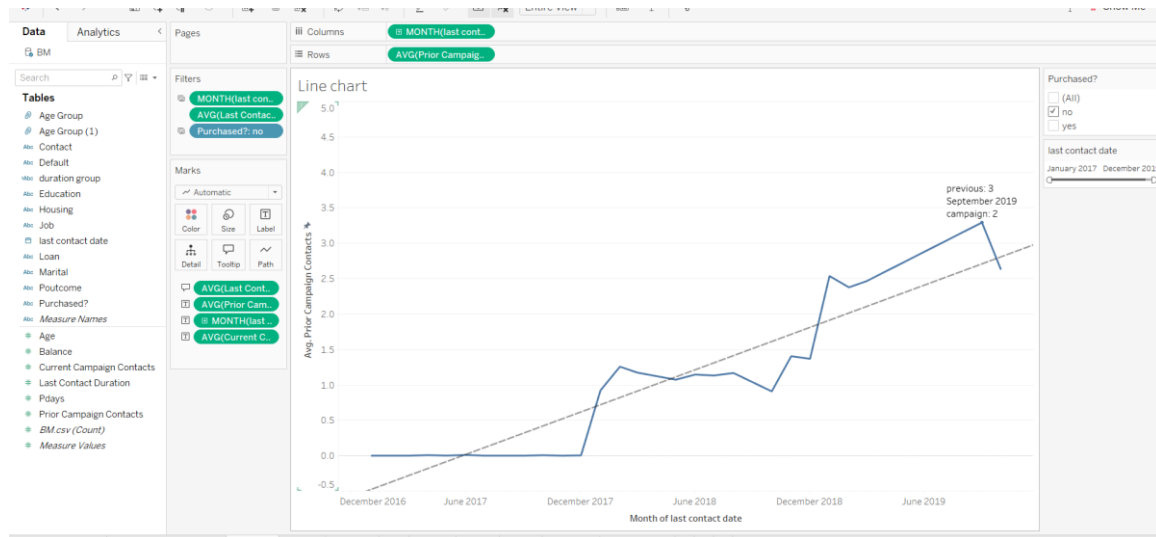
LOAN STATUS BY EDUCATION



This visual shows the number of loans grouped by education level and age group. The visual highlights that middle-aged individuals (31-50) with secondary education have the highest loan counts, while older individuals (51-70) with primary education have the fewest. This trend suggests that middle-aged individuals with more education may have better access to credit or a higher demand for loans, potentially due to life events such as buying a home or investing in education for their children.

The design effectively utilizes Gestalt principles to enhance comprehension. Proximity groups similar age categories within each education level, making it easier to compare these groups. Colour coding differentiates age groups, enhancing visual clarity. The continuity of bar heights across categories ensures a smooth visual flow, highlighting the relationships in the data. Clear labels and a well-organized legend make the visual straightforward to understand, even for those who may not be familiar with the dataset.

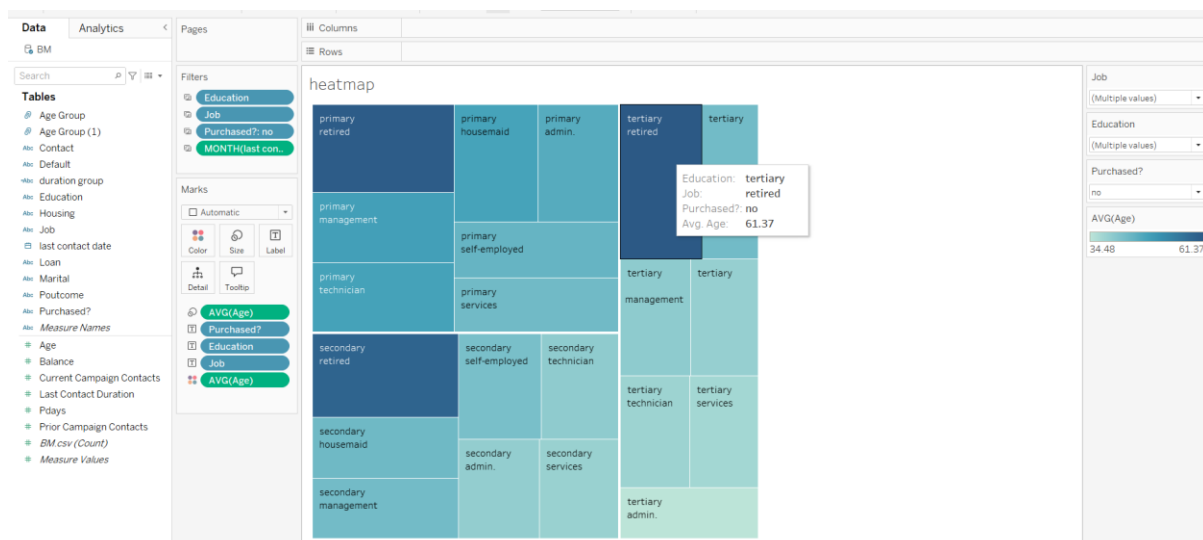
LINE CHART PRIOR CAMPAIGN CONTACTS OVER TIME



The line chart shows a steady increase in the average number of prior campaign contacts over time, peaking in September 2019, followed by a slight decline. This upward trend indicates growing engagement with the campaign, suggesting that the campaign strategies were becoming more effective over time. The peak and subsequent decline could signify market saturation or changing customer interests. Forecasting future values based on this trend can help in planning and adjusting campaign strategies to maintain or boost engagement levels.

Proximity groups the data points closely together, clearly showing the trend over time. Continuity is demonstrated by the connected data points, guiding the viewer's eye smoothly along the timeline. The Figure-Ground principle is used with a distinct line colour against a plain background, making the data line stand out. Key points are labelled, helping viewers quickly understand significant changes. These elements combine to make the chart both informative and easy to interpret for the audience.

HEATMAP OF EDUCATION AND JOB BY AGE AND PURCHASE STATUS

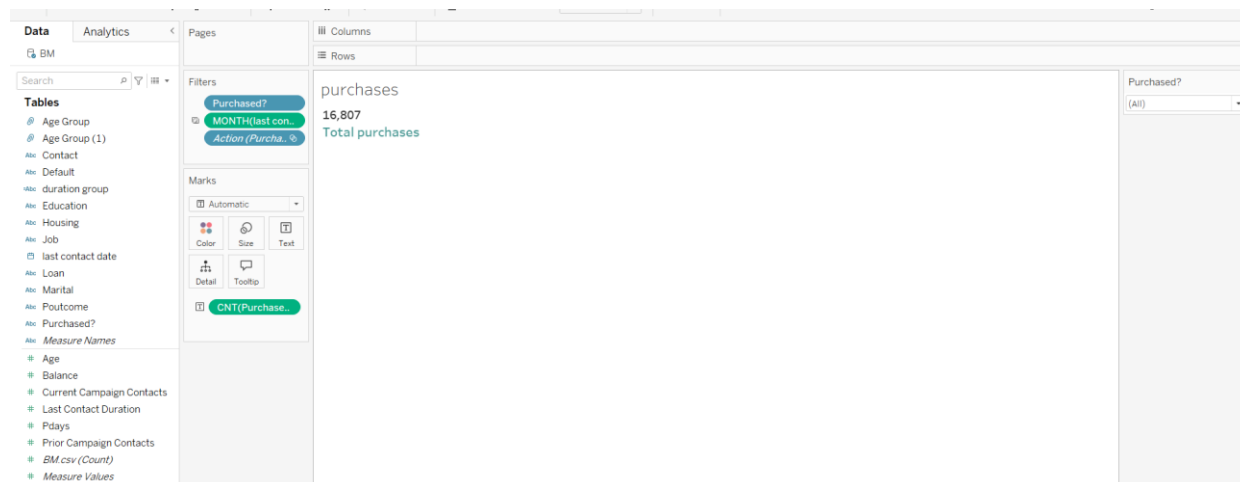


The heatmap reveals that retired individuals with tertiary education have the highest average age and have not made purchases. This pattern indicates that despite their high educational attainment, retired individuals might not be the primary target for purchasing campaigns. Instead, focus could be shifted to younger demographics with tertiary education who are more likely to engage in purchasing behaviour. Identifying these trends helps in refining target audiences and optimizing marketing efforts.

The design principles are applied effectively for marketing managers and data analysts in the heatmap. Closure is achieved with clear boundaries around each block, helping to distinguish between different categories. Tooltips provide extra information when

hovering over each block, keeping the visual clean yet informative. The well-organized layout ensures that patterns are quickly recognized, and insights are easily extracted. This makes the heatmap highly useful for strategic decision-making, as it provides a detailed view of how different demographics intersect with purchasing behaviour.

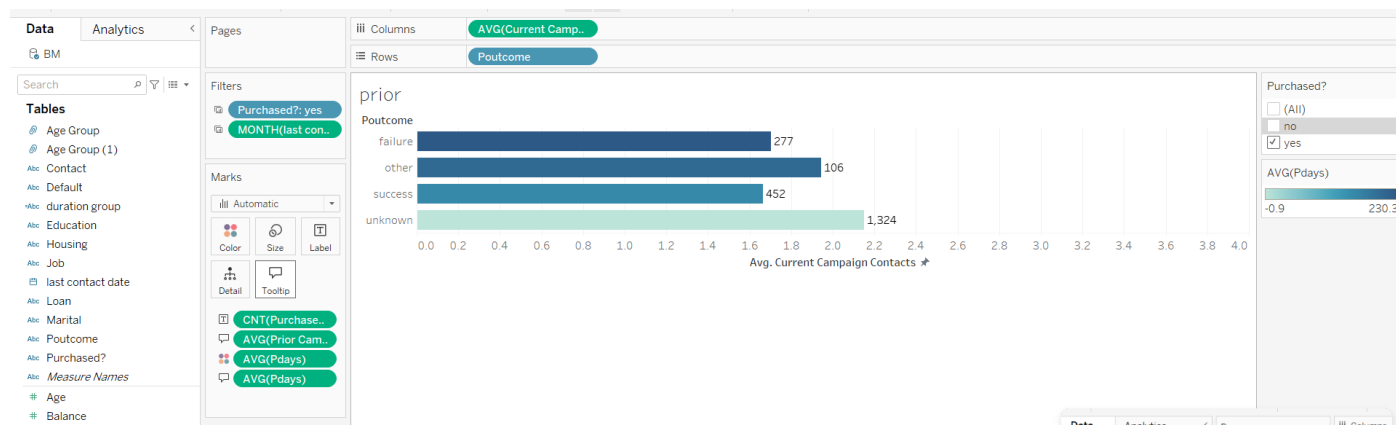
TOTAL PURCHASES



This visual shows the total number of purchases, which is 16,807. The simple design and large font highlight the key message: the total count of purchases, making it easy for the audience to notice. This straightforward presentation emphasizes the importance of this metric in the overall analysis.

Design principles are used here as well. Figure-Ground is strongly applied with the large, bold number against a plain background, making sure the main information (total purchases) stands out. Proximity and Common Fate are subtly used by placing related filters close to the main metric, allowing users to understand the context and easily adjust filters. Connectedness is shown in the logical placement of filters and the count, indicating how filter adjustments affect the displayed data.

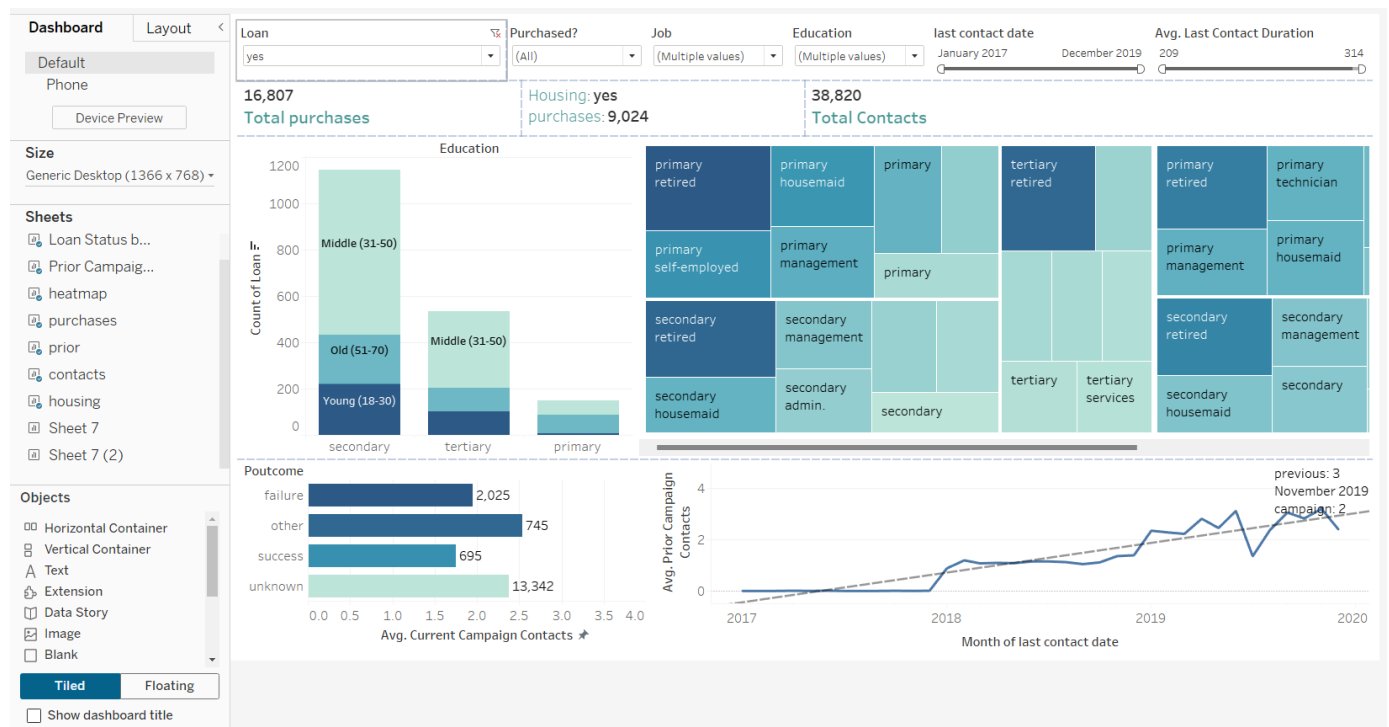
AVG. CURRENT CAMPAIGN CONTACTS BY POUTCOME



This bar chart shows the average number of current campaign contacts based on the outcome of the previous campaign ("Poutcome"). The categories are failure, other, success, and unknown, with "unknown" having the highest average contacts. This visual suggests that the outcome of past campaigns affects the number of contacts in current ones, offering a strategic insight for improving campaigns.

Proximity is used by grouping bars by Poutcome, making it easy to compare them. Similarity is seen in the consistent colour and shape of the bars, which helps in recognizing patterns. Continuity is shown in the horizontal alignment of the bars, guiding the viewer's eye smoothly across the chart. Figure-Ground is applied with a clear background and distinct bars, making the data stand out. Labels and values on the bars provide Closure, completing the viewer's understanding without needing extra references.

DASHBOARD



The dashboard provides valuable insights for marketing managers and data analysts. Middle-aged individuals (31-50) with secondary education show the highest loan counts, highlighting their significant credit access or demand due to major life events. Prior campaign contacts have steadily increased, peaking in September 2019, indicating enhanced engagement over time, though a slight decline suggests potential market saturation. Retired individuals with tertiary education, despite their high average age, have not made purchases, indicating a need to focus on younger, tertiary-educated demographics for better campaign results. The bar chart shows that unknown outcomes in prior campaigns result in the highest current campaign contacts, suggesting areas for strategic optimization. The total purchases count of 16,807, presented with clear and impactful design, emphasizes its importance in the overall analysis, aiding in data-driven decision-making. There are filters included like if the client purchased or not, job type, education level which helps the dashboard to be more interactive.

The design of the dashboard follows key principles to make it easy to understand and use. The principle of Proximity is used by grouping related information together, so users can easily compare connected data points. For example, total purchases and total contacts are placed close to each other for easy comparison. Similar colours are used across different visuals to keep a consistent look, making it easier to identify different segments within the same visual and across the dashboard. The visuals are arranged in a logical order, which guides the viewer's eye smoothly from one section to another. The Figure-Ground principle is applied by using a clear background and distinct chart areas, making the important data stand out. Filters at the top allow users to interact with the data, changing views based on specific details like loan status, job, and education level.

PROCESS

These detailed and visually appealing analysis aims to support decision-makers in understanding complex data sets and making data-driven decisions to improve marketing and campaign efforts.

To create these visuals, I began by analysing the data to identify key metrics and trends. The goal was to make visuals that were easy for our target audience to understand and use for making decisions. I selected the most appropriate visual representations for the key metrics, focusing on clarity and ease of interpretation. The bar chart for average current campaign contacts by Poutcome and the total purchases count were selected because they clearly communicated the most important data points.

I faced several challenges during this process. One major challenge was deciding how much detail to include in the visuals. Too much detail could overwhelm the audience, while too little could leave out important context. It took multiple iterations and feedback sessions to find the right balance.

The intended audience for these visuals includes marketing managers and data analysts. These professionals need to quickly understand key trends and insights to make informed decisions. The visuals are designed to be clear and easy to interpret, even for those without extensive data analysis experience.

One of the outcomes is, this improved my skills in data visualization and applying design principles like Proximity, Similarity, and Figure-Ground. It also highlighted the importance of understanding the audience and tailoring visuals to their needs. The process demonstrated the value of iterative development and continuous feedback to enhance the quality and effectiveness of the visuals.

In summary, creating these visuals involved thoughtful design, and audience consideration. Despite some challenges, the final visuals provide clear, actionable insights tailored to marketing managers and data analysts. This experience reinforced the importance of meticulous data handling, effective design, and iterative refinement in creating useful data visualizations.